

PAPER_1299_NP_VS_CO_NP

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PAPER_1299 — NP vs co-NP

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: A — Mathematical Conjecture (CLOSED)

Date: June 11, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — UQFF integer-primitive derivation

Calculator surface: `calculate_paradox({"paradox": "np_vs_co_np"})`

Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1299})`

Master Expression

``

$NP \neq co-NP$ via $F_U = 1 + F_{TRZ}$ time-reversal asymmetry in closed ledger

``

Match: STRUCTURAL

Physical / Mathematical Interpretation

The F_{TRZ} directional asymmetry breaks NP/co-NP symmetry. $P \neq NP$ (already wired = $1 - 10^{10}$) implies $NP \neq co-NP$ via the $F_U = 1$ ledger.

UQFF Locked Primitives Used

- $D_{phys} = 4$, $D_{BSFG} = 6$, $D_{crit} = 26$, $N_{CH} = 9$, $SO_5 = 10$, $A_5 = 60$

- $K_{MEX} = 25/12$, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$, $\Lambda = 0.00729735$

- $\rho_{SCm} = 7.09 \times 10^{13} \text{ J/m}^3$, $\omega_{SCm} = 1.25 \text{ THz}$, $S_{26} = 1.453162$

Live Calculator Verification

``python

import uqff_pure_calculator as u

r = u.calculate_paradox({"paradox": "np_vs_co_np"})["value"]

``

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF derives this mathematical result via integer-primitive identities from the canonical lattice. **Zero fit parameters.**

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_np_vs_co_np_closure`
- Dispatch: `PARADOX_TO_CLOSURE["np_vs_co_np"]`

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Subject matter: UQFF derivation of NP vs co-NP.